Statistical Foundations of Latent Structure and Representation Learning

Early-Career Researcher Poster Session

Wednesday, November 4, 2026 | 4:45–6:15 p.m.
Room 214, School of Social Work, Columbia University
1255 Amsterdam Avenue, New York, NY 10027
Presenters listed alphabetically by surname.

Ozgur (Oz) Beker
Columbia University
DisCoLoRA: Variational Disentanglement of Pretrained Representations

Jinhang Chai
Princeton University
Deep Transfer Q-Learning for Offline Non-Stationary Reinforcement Learning

Abhinav Chakraborty
Columbia University
The Geometry of Multitask Learning: Sharp Rates for Shared Subspace Estimation

Tianyi Chen
Johns Hopkins University
Vertex misalignment and changepoint localization in network time series

Chengyu Cui
University of Michigan
Marginal maximum likelihood estimation and asymptotic theory for latent variable models in high dimensions

Mia Dai
Columbia University
Sparse-View Interpretable 3D Animal Behavior Representations for Neural Encoding and Decoding

Wilson G. Gregory
Columbia University
Warm-starting PDE solvers with any-dimensional machine learning

John Hood
University of Chicago
Near-Universal Multiplicative Updates for Nonnegative Einsum Factorization

Chengzhu Huang
Columbia University
From Good Starts to Optimal Inference: Generalized Latent Factor Models with Missingness and Implicit Regularization

Yoonhaeng Hur
Columbia University
Inference for Clustering: Conformal Sets for Cluster Labels

Seunghyun Lee
Harvard University
High-dimensional latent class models beyond local independence

Mengqi Lin
University of Michigan
Characterizing Identifiability in Boolean Factor Models

Xindi Lin
Yale University
Who Benefits from Network Interventions? Debiased Machine Learning of Heterogeneous Causal Effects under Network Interference

Jia Liu
Columbia University
Scalable Text-Embedding-informed Cognitive Diagnosis of Large Language Models

Luigi Maria Malgieri
Duke University
DIRECT: Dirichlet Coupling for Multiview Clustering

Lorenzo Mauri
Duke University
Overfitted high-dimensional matrix factorizations via adaptive spectral shrinkage

Yaroslav Mukhin
Cornell University
Kernel von Mises Formula of the Influence Function
Surrogate-powered Causal Inference on Censored Outcomes

Kieran Murphy
New Jersey Institute of Technology
Add latent stochasticity to track information flow through transformers

Michael Pressler
Columbia University
Cell-JEPA: Self-Supervised Representation Learning for Cell Trajectories

Roshni Sahoo
Columbia University
Data Fusion for High-Resolution Estimation

Heejune Sheen

Yale University

Statistical Analysis of Synthetic Data and Verification for Fine-Tuning: A Linear Regression Framework

Frank Shih

Memorial Sloan Kettering Cancer Center

Variable Selection and Uncertainty Quantification in High-Dimensional and Constrained Cox Models via Extended Fiducial Inference

David Snider

Columbia University

Mixed Membership Model of Low-rank Matrices with Multimodal Extension

Runshi Tang

Columbia University

RECAP: Representation-Based Calibration with Amortized Pooling for Frequentist Simulation-Based Inference

Md Kaosar Uddin

Auburn University

Interpretable Domain Generalization for Multi-Site Functional Connectivity Analysis

Emilie Wedenborg

Columbia University, Arts and Sciences

The Polytopal Neural Network

Zhiyu Xu

Columbia University

Latency-Response Theory Model: Evaluating Large Language Models via Response Accuracy and Chain-of-Thought Length

Haochen Yang

Columbia University

Latent Representation Learning for Quantifying Morphological Variability in Developing Mouse Gut Tube

Qishuo Yin

Princeton University

SMART Fine-tuning Factor Augmented Neural Lasso

Wenjin Zhang

Columbia University

Discrete Causal Representation Learning